

A further conjecture on OEIS A239033, settled by the proof already in hand

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Abstract

OEIS A239033 carries more than one conjecture. One of them has a proof; this paper settles a DIFFERENT one, and it needs no new model and no new engine – only the linear recurrence already established for the sequence. A claim about a C-finite sequence whose own characteristic polynomial is a multiple of the established one is decided by polynomial divisibility, and where the claim is an identity rather than a recurrence, by divisibility together with a finite window of agreement.

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1 The premise

The entry's first terms are 8, 43, 212, 806, 2592, 7265, 18362, 42809, ..., with offset 1. The sequence satisfies

$$a(n) = 10a(n-1) - 45a(n-2) + 120a(n-3) - 210a(n-4) + 252a(n-5) - 210a(n-6) + 120a(n-7) - 45a(n-8) + 10a(n-9)$$

a linear recurrence of order 10 with characteristic polynomial

$$q(x) = x^{10} - 10x^9 + 45x^8 - 120x^7 + 210x^6 - 252x^5 + 210x^4 - 120x^3 + 45x^2 - 10x + 1.$$

This recurrence is proved in this project; it is the premise of everything below and nothing else is assumed. It was confirmed against every term the entry publishes from the index where it begins to hold, with at least 10 + 2 further terms after that index.

2 The further conjecture

Separately from the conjecture already settled, the entry records

$$\begin{aligned} \text{Empirical: } a(n) = & (1/8640)*n^9 - (61/20160)*n^8 + (263/3360)*n^7 - \\ & (1439/1440)*n^6 + (29843/2880)*n^5 - (194227/2880)*n^4 + (668597/2160)*n^3 - \\ & (1477657/1680)*n^2 + (59263/42)*n - 943 \text{ for } n > 2. \end{aligned}$$

The further claim is a CLOSED FORM, and for that the divisibility is not the whole argument. It gives that a and the closed form f satisfy the same linear recurrence; it does not say they are the same solution of it. Write p for the annihilator of f ,

$$p(x) = x^{10} - 10x^9 + 45x^8 - 120x^7 + 210x^6 - 252x^5 + 210x^4 - 120x^3 + 45x^2 - 10x + 1,$$

of order $D = 10$. Since $q \mid p$ and a satisfies the recurrence of q , a satisfies that of p as well; f satisfies it by construction. Two solutions of a linear recurrence of order D with nonzero leading coefficient agree everywhere as soon as they agree at D consecutive indices, and a and f were checked to agree at $n = 13, \dots, 22$, which lies inside the terms the entry publishes and past the index from which a provably satisfies p . Hence $a(n) = f(n)$ for every n in the claimed range.

Theorem 1. *The conjecture displayed above is true.*

3 What is and is not claimed here

This is a second result on an entry that already has one, and it is a second result only because it is a DIFFERENT statement. A generating function whose denominator's reciprocal is exactly q says precisely what the proved recurrence says, in another notation; such restatements are identified and are not counted. The conjecture settled here is not of that kind.